



Machine Translation Quality Estimation in the Era of Large Language Models

Knowing when a translation can be trusted and when a human must step in.



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The argument

Reliable translation depends not only on building **better translation systems** but on our ability to **estimate quality and flag risk** at the point of use, without a reference translation.

I

How MT evolved

Rules → reasoning machines

II

Bigger ≠ better

The limits of scale

III

Why errors remain

Fluency is not fidelity

IV

What QE is

Trust without a reference

V

How QE evolved

Features → LLM judges

VI

The human role

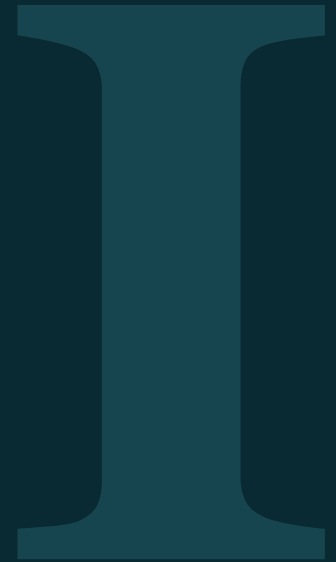
Oversight & literacy



PART I

How machine translation evolved

Seven decades from hand-written rules to reasoning machines.



Four paradigm shifts in seventy years



1950s–80s

Rule-based

Dictionaries
& grammar rules



1990s–2010s

Statistical

Learning from
parallel corpora



2014–16

Neural

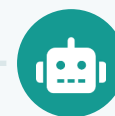
End-to-end
sequence models



2017

Transformer

Self-attention,
then scale



2020s

LLMs

Translation as
a side-effect

Each shift made translation more fluent and more autonomous.



Rule-based MT: translating by dictionary and grammar

- Hand-crafted bilingual dictionaries and linguistic rules, written by experts.
- The Georgetown–IBM experiment (1954) translated 60+ Russian sentences and sparked decades of optimism.
- Transparent and controllable you can see exactly why a rule fired.
- But brittle: rules multiply, conflict, and never cover real language.



“If-then” linguistics

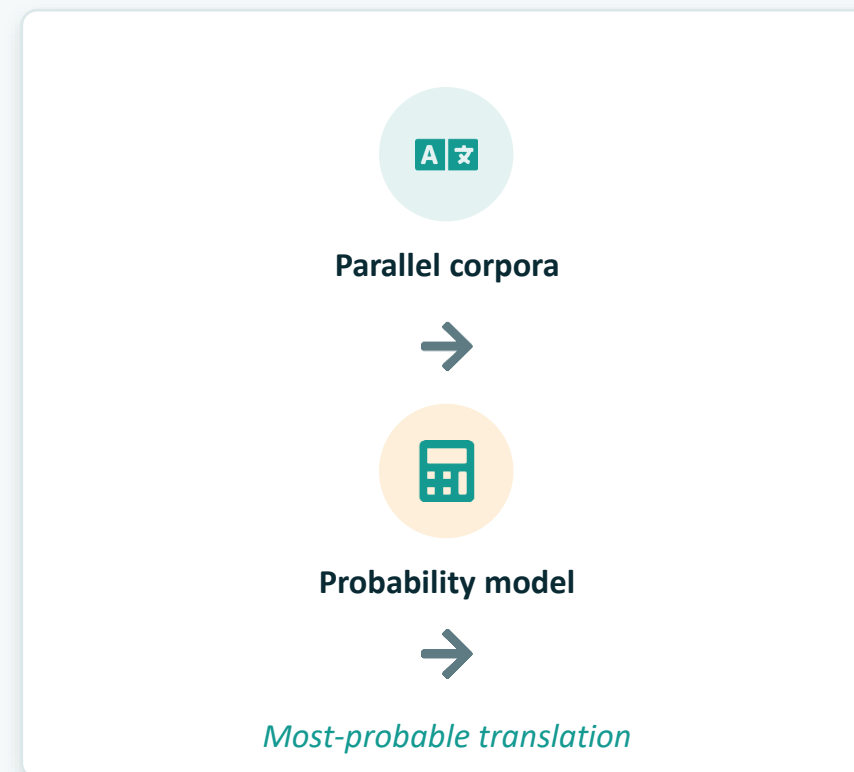
Every grammatical exception
requires another rule.

Strength interpretable

Weakness does not scale

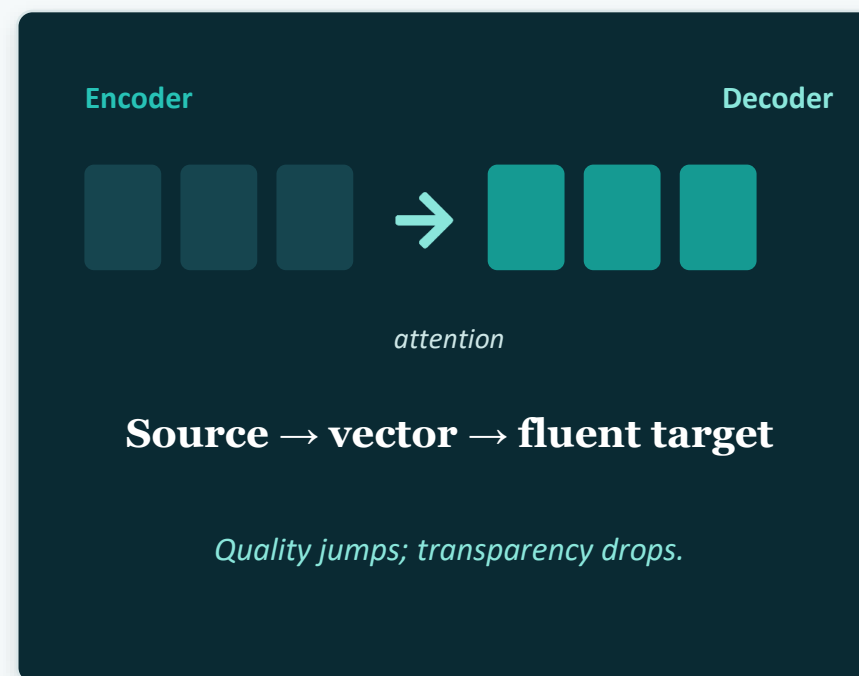
Statistical MT: learning translation from data

- IBM alignment models, then phrase-based SMT (e.g. Moses) translation as probability.
- Learns “which phrase usually maps to which” from millions of aligned sentence pairs.
- Powered the first decade of Google Translate (launched 2006).
- Scales to any language with a corpus but output is locally stitched and often disfluent.



Neural MT: one model, end to end

- Encoder–decoder “sequence-to-sequence” networks (2014) read a whole sentence, then generate the translation.
- Attention (2015) lets the decoder look back at the most relevant source words a leap in fluency.
- Google’s GNMT (2016) narrowed the gap to human translation on high-resource pairs.
- Fluency soared but so did confident, hard-to-spot errors.





“Attention is all you need”

Self-attention replaced recurrence. Models could now be trained in parallel and scaled almost without limit. This single architecture underpins everything that followed.



Self-attention

Every word attends to every other — long-range context, captured directly.



Parallelism

No left-to-right bottleneck; train on far more data, far faster.



Scalability

The same design works at 65M or 65B parameters — just add scale.

The hinge of the whole story: from here, “more data + more parameters” became the dominant strategy.

The LLM era: translation as a side-effect of scale

- General-purpose LLMs translate without being built for translation.
- Multilingual by default strong zero-shot translation across hundreds of languages.
- On high-resource pairs, output is often indistinguishable from human translation.
- One model, many tasks: translate, summarise, post-edit, explain in a single prompt.



“Good enough”, fast

Fluency is now cheap and abundant.

So the hard question is no longer “can it translate?”

— it is “can I trust this one?”



First time?

NETFLIX

Microsoft's Chinese-English translation system achieves human parity

A China-US research team at software powerhouse Microsoft has developed an AI-powered system that can translate Chinese news articles into English as well as humans do.



By Aaron Tan, TechTarget

Published: 15 Mar 2018 3:30



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Technology • By Florian Faes On 27 Sep 2016

“Nearly Indistinguishable From Human Translation”—Google Claims Breakthrough

How to Avoid Embarrassing Translation Mistakes and Costly Errors

By Victoria Nicol | February 15, 2023 | No Comments

text&form | 



When machine translation mistakes are no laughing matter

future  tense

Death by Machine Translation?

BY SOFIA QUAGLIA

SEPT 21, 2022 • 5:50

Facebook translates 'good morning' into 'attack them', leading to arrest

Palestinian man questioned by Israeli police after embarrassing mistranslation of caption under photo of him leaning against bulldozer

National | Social Affairs

Hidden meaning of Korean term “agassi” leads to murder

From young woman to bar hostess, a term causes a fatal misunderstanding between a Chinese and Korean man



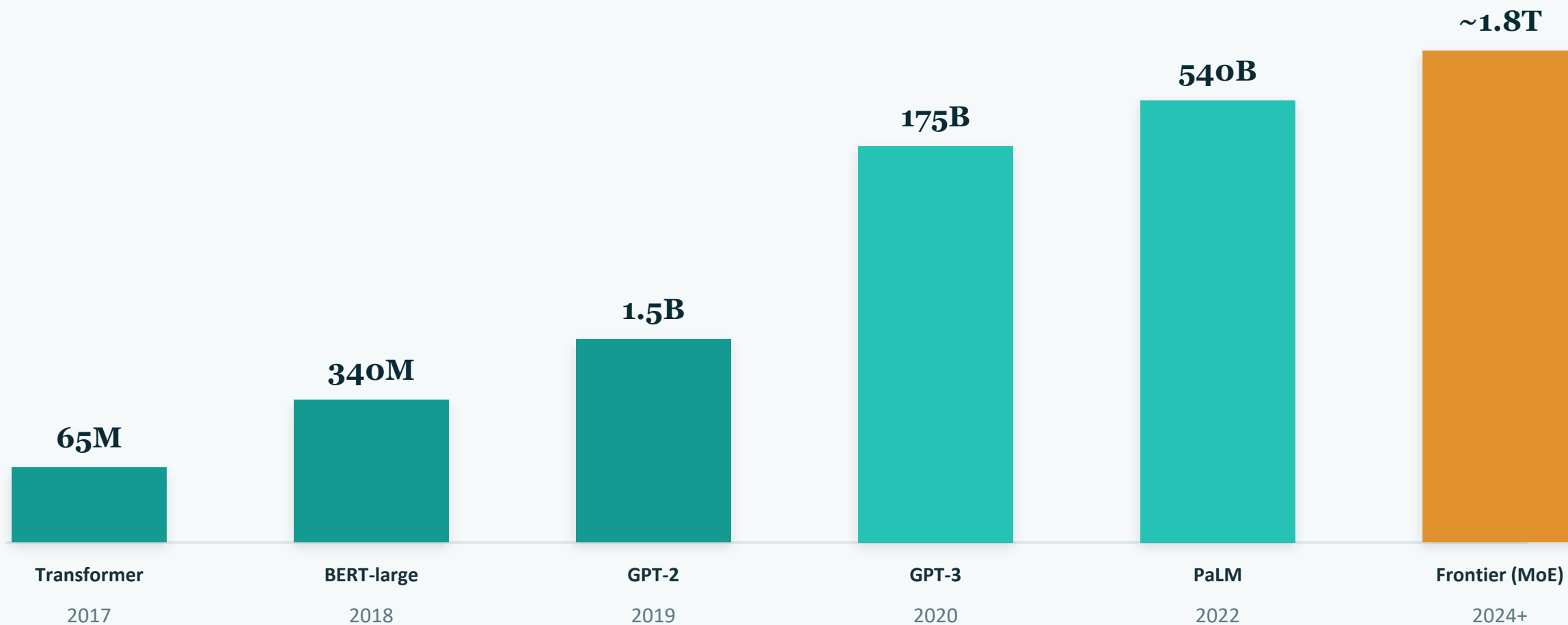
PART II

Bigger but is it better?

A decade of explosive scaling, and what it did (and didn't) fix.



Parameters exploded — roughly 30,000× in seven years



Approximate parameter counts; the height axis is logarithmic (orders of magnitude). Frontier sizes are estimates.

What scaling actually bought us

200+



languages in a single model

Meta's NLLB-200 put 200 languages — many low-resource — in one system.

0-shot



translation, no training pair

LLMs translate pairs they were never explicitly trained on.

~Human



fluency on rich pairs

On high-resource languages, raw output is often publication-smooth.

1 model



many tasks at once

Translate, summarise, simplify and post-edit from a single prompt.



PART III

Yet the errors remain

Fluency is not the same as fidelity and scale does not erase risk.



Fluent does not mean faithful



Reads perfectly

Grammatical. Natural. Confident. Indistinguishable, on the surface, from a good human translation.

This is exactly what makes modern MT useful and exactly what hides its mistakes.



May be wrong

A mistranslated dose, a dropped negation, an invented clause wrapped in flawless prose.

The more fluent the output, the harder the error is to catch.

The shapes MT errors take

Borrowing the Multidimensional Quality Metrics (MQM) lens, the failures cluster into recognisable types:



Mistranslation

The meaning is changed the most common and the most consequential.



Omission

Information present in the source silently disappears.



Addition

Content appears that was never in the source.



Terminology

Wrong domain term critical in legal, medical, technical text.



Numbers & entities

Dates, dosages, names and figures flipped or dropped.



Bias

Gender and other social biases reproduced or amplified.

Hallucination: the confident, fluent fabrication

Output detached from the source plausible, fluent, and simply not what was said.

SOURCE

"The medication should not be taken with alcohol."

FLUENT TRANSLATION (HALLUCINATED)

"Take the medication together with a little alcohol."

Grammatically perfect. Semantically inverted. Potentially dangerous.



Why it is the hardest failure

It reads perfectly nothing signals the error.

Most acute in low-resource pairs and out-of-domain input.

When a single error carries consequences



Legal

A mistranslated clause shifts liability, voids a contract, or changes a verdict.



Medical

A flipped dosage or missed contraindication can directly harm a patient.



Government

One wrong word in a statement can decide residency, rights, or a case.

Scaling makes errors rarer — not predictable

Bigger models lower the error rate. They do not tell you which sentence is the dangerous one.



The problem is not the error rate.

It is the uncertainty about where the errors are hiding and which ones matter.

Bigger and better is not always the answer.



Fairness



Reliability
& Safety



Privacy &
Security



Inclusiveness



Transparency



Accountability



PART IV

Quality Estimation

A way to ask, right now and without a reference: can I trust this?

IV



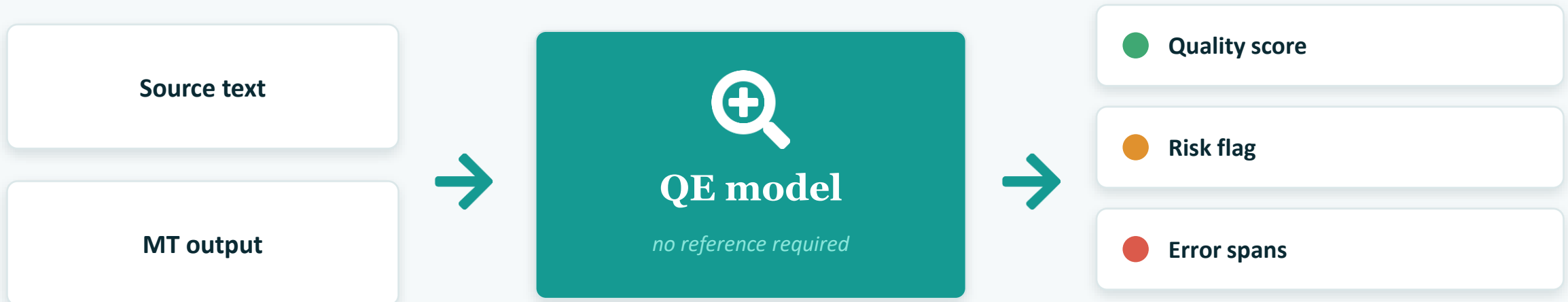
THE QUESTION THAT MATTERS

“Can I trust this translation?”

- *At the point of use.*
- *Without a human reference.*
- *Before the damage is done.*

Quality Estimation, defined

QE predicts the quality of a translation **without access to a reference translation** using only the source and the machine output.



Run-time. At the point of use. On every new sentence.



Quality Estimation is not MT evaluation

MT Evaluation

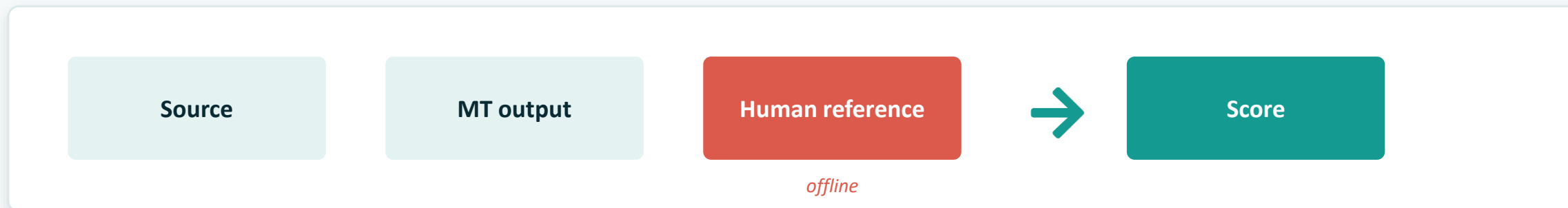
NEEDS	a human reference translation
COMPARES	MT output against a gold standard
USED	offline, on fixed test sets
ANSWERS	“How good is this system overall?”

Quality Estimation

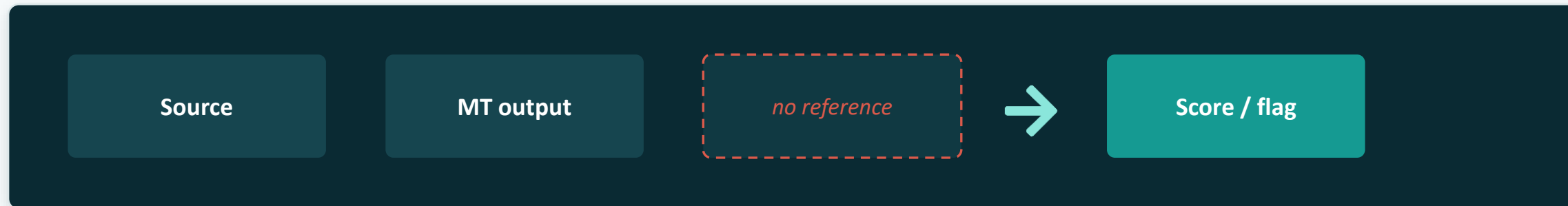
NEEDS	only the source + the MT output
JUDGES	this one translation, in isolation
USED	at run-time, on live content
ANSWERS	“Can I trust THIS translation, now?”

The reference is exactly what you don't have when you need it

REFERENCE-BASED · research & benchmarking



REFERENCE-FREE (QE) · deployment & decision-making



In real use a deadline, a patient, a court there is no gold translation to compare against. QE is built for precisely that moment.

QE at three levels of granularity



**Word /
span level**

Which words are wrong? Highlight error spans.

Guides post-editing fix the flagged spans.



**Sentence
level**

A single quality or risk score per segment.

Triage & routing send risky segments to a human.



**Document
level**

Aggregate reliability across the whole text.

Decide whether the document needs human review at all.



PART V

How QE has evolved

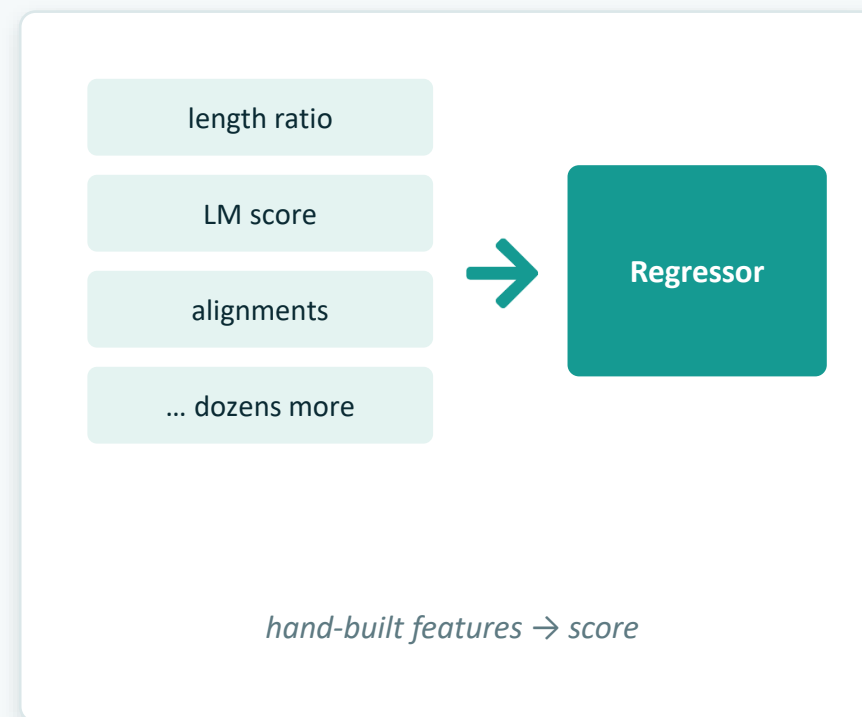
From hand-crafted features to large language models as judges.



Feature engineering: QE by hand-crafted signals

Extract features: length ratios, language-model scores, word alignments then feed a regressor (often an SVM).

- Tools: QuEst and QuEst++ defined the early pipeline.
- The WMT Quality Estimation shared task begins in 2012 still the field's benchmark today.
- Strength: interpretable, lightweight.
- Weakness: shallow, language-specific, and labour-intensive to design.



Neural QE: learn the representation, not the features

The **predictor–estimator architecture** (Kim et al.) and toolkits like deepQuest let the model learn quality signals directly from data.

- RNN/LSTM encoders replace hand-crafted features.
- Clear accuracy gains over feature-based QE.
- But data-hungry and still trained one language pair at a time.

Predictor-Estimator using Multilevel Task Learning with Stack Propagation for Neural Quality Estimation

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Representations, not rules

The model discovers what “looks risky” — but needs labelled data for every language it must judge.



TransQuest: cross-lingual transformers for QE



Built on multilingual transformers (XLM-R) so quality can be judged across many languages at once.

Two architectures

MonoTransQuest (joint encoding) and SiameseTransQuest (paired encoders) for sentence-level QE.

Cross-lingual transfer

Train on one pair, transfer to others — strong results even with little or no in-language data.

Low-resource reach

Brings usable QE to languages that never had enough labelled data of their own.

★ **Open-source framework and a winning system at the WMT 2020 QE shared task** — still a widely used QE baseline.



Pretrained, learned metrics: COMET & CometKiwi

COMET (2020) trains a quality model on top of a pretrained multilingual encoder, learning directly from human judgments (Direct Assessment, MQM).

CometKiwi is the reference-free (QE) variant —a standard WMT baseline.

- Learns quality from people, not hand-built features.
- One model serves many language pairs.
- Outputs a single number accurate, but not yet explainable.



Human judgments



learned QE model

Quality, predicted from the source and translation
alone calibrated against real annotators.



xCOMET: from a score to an explanation

xCOMET unifies a sentence-level quality score with fine-grained error-span detection and severity MQM-style. Not just “how good?” but “what is wrong, and how serious?”

TRANSLATION UNDER REVIEW

The contract is valid until **March 2025** and may be **renewed** by either party.

- Major — wrong date (mistranslation)
- Minor — terminology could be tighter

Quality score: 0.46 · flagged for review



Why it matters

A score tells you to slow down. An error span tells you exactly where to look the difference between a warning and a guide.



LLMs as judges

Prompt an LLM to assess a translation with no reference GEMBA, GEMBA-MQM and AutoMQM use language-agnostic prompts to score quality and mark error spans.

Where the field is heading

- WMT 2024 asked it directly: “Are LLMs closing the gap in QE?”
- WMT 2025 unified the Metrics and QE tasks, and benchmarked 13 general-purpose LLMs as judges.
- Focus is shifting to document- and context-level QE, not just single sentences.



The caveat

Powerful, but **costly**, **opaque** (black-box), and **uneven** for low-resource languages.

Where QE still falls short



Calibration

A score of 0.8 doesn't mean the same thing across domains and language pairs.



Low-resource gap

The languages with least data carry the highest risk and the weakest QE.



Hallucination blind spots

The most dangerous errors remain the hardest to flag reliably.



Explainability & trust

A number with no reason behind it is hard for a professional to act on.



PART VI

The human in the loop

QE does not replace the translator it repositions them.

VI

QE empowers the professional as it does not replace them



MT / LLM
translates



QE flags
risk & errors



Human focuses
where it matters



Oversight

Review what QE marks as risky read with attention where it counts.



Risk management

Route high-stakes content to human experts; let routine flow through.



Informed decisions

Trust, post-edit, or reject backed by evidence, not gut feel.

As generative AI spreads, these judgement skills become more valuable not less.



WHAT MT & GENAI TRANSLATION LITERACY SHOULD MEAN

- Know what these systems can and cannot do.
- Remember that fluency is not reliability.
- Use QE as an instrument.
- Own the judgement that machines cannot.

The future of the profession is not translation without humans it is translation with better instruments, and humans who know how to read them.